

Chinmay Kapoor

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AI Engineer | Agentic AI Platforms | Edge AI/ML Systems | Data Science & Applied ML

SUMMARY

AI/ML engineer with 3+ years building applied AI systems across agentic AI, LLM/RAG platforms, embedded Edge AI, data science, forecasting, and production-oriented ML workflows. Built end-to-end AI platforms from product vision through backend architecture, model/runtime integration, governed tool execution, observability, deployment guidance, demos, and customer enablement. Strong in Python, FastAPI, JavaScript/TypeScript, C/C++, SQL, local/cloud LLM routing, vector search, GPU/accelerator-aware inference, data pipelines, evaluation, and cross-functional delivery with software, hardware, sales, data, and customer teams.

CORE AI ENGINEERING STRENGTHS

- **Agentic AI platforms:** Built multi-agent orchestration, session/state management, model-provider routing, tool/function calling, MCP-style tool exposure, ADK-compatible runtime paths, governed file/search/shell/API actions, audit logs, and traceable execution workflows.
- **Production GenAI systems:** Built RAG interfaces, local/cloud LLM integrations, vector search, OpenAI/Gemini-style API paths, model fallback, streaming APIs, evaluation loops, prompt/tool-call debugging, and production-readiness documentation.
- **Edge AI and accelerator deployment:** Validated AI workloads across Linux, ARM, x86, GPU/NPU/CPU setups, embedded devices, NVIDIA Ampere-class edge GPUs, AMD Zen x86 APUs with Radeon iGPUs, Vulkan-backed llama.cpp, CUDA/ROCm-aware environments, Hailo/RK3588 accelerators and Zynq UltraScale+ MPSoC platforms.
- **Data science and ML systems:** Built forecasting, classification, computer-vision, telemetry analytics, benchmarking, feature engineering, quantization, ETL, and model evaluation workflows across enterprise, environmental, financial, and sensor datasets.

SIGNATURE PLATFORM: KAPTAIN[†]

- **End-to-end ownership:** Built Kaptain as a full AI-agent control and development platform, covering product direction, user workflows, backend architecture, model-provider routing, local runtime setup, governed tool execution, code intelligence, observability, documentation, and demo enablement.
- **Agent orchestration layer:** Designed the runtime for multi-turn agent sessions with streaming responses, session history, provider fallback, local/cloud model routing, CLI-backed model support, OpenAI-compatible endpoints, Ollama/local model paths, and web/mobile parity.
- **Governed execution broker:** Built the bridge between model intent and real system actions by turning tool calls into typed file, search, shell, API, planning, and edit operations with workspace policy, risk checks, approval flows, audit logs, and traceable results.
- **Code intelligence and context:** Built tree-sitter and SQLite-backed indexing so the platform could reason over APIs, functions, files, dependencies, symbols, and project structure before making code or configuration changes.
- **Blackbox / God's Eye observability:** Built trace and dependency views for API calls, session control, tool execution, model rounds, context injection, service dependencies, latency points, bottlenecks, and failure paths so agent behavior could be inspected and explained.
- **Local model and hardware-aware setup:** Built environment-aware setup services for GGUF discovery/download, llama.cpp/llama-server setup, Ollama/local tool detection, GPU backend detection, GPU-layer controls, fallback handling, and runtime readiness across Linux, ARM, x86, and edge devices.
- **MCP, custom agents, and team workflows:** Worked across MCP-style tool exposure, custom-agent connection flows, delegate routing, team-agent concepts, capability-aware execution, and reusable action contracts so tools and agents could interoperate through one controlled interface.
- **AI product engineering:** Balanced powerful autonomous-agent behavior with customer-facing requirements for safety, explainability, setup reliability, local/cloud flexibility, deployment diagnostics, and field-demo stability.

EXPERIENCE

Field Application Engineer - AI Specialist, MIMIK (Vancouver, BC)

Oct 2025 - Present

- **MimOE Studio:** Built MimOE Studio for MIMIK by productizing Kaptain's agent-control capabilities into customer-facing AI workflows, shaping demo paths, field controls, setup flows, runtime guidance, and deployment narratives for internal teams and external audiences.

- Worked on context injection, model handling, and agentic development kit integration so connected agents could preserve session state, route across local/cloud providers, use external tools consistently, and deliver a more polished customer experience.
- Turned platform internals into field-ready experiences by exposing model selection, session continuity, local/cloud routing, runtime setup, and debugging views through interfaces that sales, engineers, and technical customers could operate during demos.
- Reduced setup friction for MimOE Studio demos by packaging environment-aware runtime checks for local model tools, GPU backends, GGUF scan/download, llama.cpp/llama-server/Ollama setup paths, and GPU-layer controls across Linux, ARM, x86, and edge systems.
- Supported customer-facing agent demos with Blackbox-style trace views and d3.js service maps, giving field teams enough backend visibility to explain API flow, session behavior, runtime bottlenecks, and failure paths without exposing unnecessary implementation detail.
- Implemented Vulkan-backed llama.cpp paths on AMD iGPUs for portable local inference when ROCm was not the practical deployment path, tuning GPU-layer offload and CPU fallback for shared-memory limits, latency, and thermal/power behavior.
- Built customer-facing Edge AI demonstrations for robotics and incident-response workflows across Arm SBCs, NVIDIA Ampere-class edge GPUs, AMD Zen x86 APU/Radeon iGPU systems, and AMD Zynq UltraScale+ MPSoC/Kria-class devices using ROS2, computer vision, speech/audio recognition, TTS, REST services, and local agents.
- Developed Python watcher services and BLE signal-correlation pipelines for beacon, gateway, tag, and location-triggered events; tested missed-location, false-zone, delayed-response, and cascading-failure cases to improve incident-handling accuracy by 30%.
- Created deployment guides, user demos, architecture walkthroughs, domain videos, and product demonstrations used online, at events, and on MIMIK social channels, translating backend implementation knowledge into customer-ready enablement.

Data Scientist - Lead / AI Consultant, *Swissvault Global (Vancouver, BC)*

Jan 2024 - Oct 2025

- **COMPIO**: Built a client-facing benchmarking and workload-simulation platform that replayed real customer workflows from API/web ingestion, object storage, and retrieval through RAG and AI inference, helping customers choose cloud, on-prem, edge, or hybrid AI architectures.
- Built integrations across VaultFS, AWS S3, GCS, Azure Blob, Azure/Fabric, local LLMs, OpenAI/Gemini-style APIs, Ollama, Hugging Face, Qdrant, ChromaDB, and FAISS, creating governed retrieval and inference paths for customer-controlled data environments.
- **SVSI**: Developed a secure local-first RAG interface for VaultFS so business users could query sensitive enterprise files through local LLMs, with optional cloud API offload or anonymizer layers for non-sensitive workloads.
- Reduced edge inference latency and lowered OPEX by 21% by evaluating retrieval quality, quantization tradeoffs, model/runtime fit, and offload decisions across CPU/GPU/NPU setups, then applying INT4/INT8 and local/cloud routing strategies.
- Collaborated with hardware and software teams to develop ML benchmarks and performance reports across NPU/GPU/CPU configurations, including on-prem/edge systems, NVIDIA GPUs, Hailo, RK3588/RKNN, CPUs, and AWS EC2/Trainium-backed deployments.
- Built batch and streaming ETL pipelines for semi-structured telemetry and system-performance data using Microsoft Fabric, Databricks, ADLS2, PySpark, SQL, Grafana, and Prometheus, improving transformation reliability and reducing load/processing time by over 30%.
- Designed forecasting and predictive-maintenance workflows for storage and system-performance use cases, using temporal features, lag behavior, rolling-window patterns, trend detection, and hardware-health indicators for operational monitoring.
- Built Imagilect computer-vision retrieval features using YOLOv5, DeepSORT, OCR, GStreamer, OpenCV, and ResNet50, linking model outputs to VaultFS-backed structured storage so users could search and analyze video-derived records.
- Accelerated POC-to-MVP delivery by 45% by translating client requirements into scoped features with software, hardware, sales, and data engineering teams, supported by whitepapers, FAQs, benchmark reports, performance reports, and technical documentation.
- Represented the organization at industry and networking events including VSW 2025 and Web Summit 2025, moving opportunities from early technical conversations into POC/MVP planning and strategic partnership discussions.

Research Assistant (ML Team Lead), *SOLIDS Lab, UVIC (Victoria, BC)*

Sep 2021 - Aug 2024

- Developed **FUSTT**, a spatio-temporal transformer for long-term multivariate forecasting on real-world marine sensor datasets for an NRC-sponsored digital twin initiative, improving forecasting accuracy by 16% and compute time by 19%.
- Designed, trained, and validated forecasting pipelines using spatio-temporal Transformers, traditional time-series Transformers, DLinear, CNN-LSTM, LSTM-based baselines, ablation studies, temporal feature engineering, missing-data handling, linear interpolation, and MSE/MAE evaluation.

- Built scalable Python workflows for preprocessing, EDA, model training, validation, baseline comparison, error analysis, and interpretation across large environmental datasets, contributing to deployment-oriented forecasting research for ecological decision-support systems.
- Applied neural networks and classical ML methods to environmental, material, and sensor-based datasets, including forecasting, classification, physicochemical property prediction, model interpretation, and decision-support analysis.
- Worked with domain experts and research collaborators to interpret model outputs, validate forecasting behavior, and connect predictive results to ecological monitoring, Blue Carbon, and climate-resilience use cases.
- Published 3 peer-reviewed works across IEEE and Springer venues, filed or received 3 grants through UVic/NRC/Mitacs, presented at CSCE 2022 and UVic/OCEAN CleanTech events, and served as mentor/co-judge for OCEAN data-driven innovation challenges.

SKILLS

Agentic AI & LLM Platforms : multi-agent workflows, Kaptain, MimOE Studio, Model Context Protocol (MCP)-style tool integration, ADK-compatible runtime work, tool/function calling, governed tool execution, approval workflows, action brokers, policy engines, ReAct-style loops, state/session management, provider routing, context injection, audit logs, JSONL tracing, LangChain, LangGraph, LangSmith, PydanticAI

LLM, RAG & Model Serving : RAG, Graph-RAG, embeddings, retrieval evaluation, vector search, prompt/tool-call debugging, Qdrant, ChromaDB, FAISS, Ollama, Hugging Face, GGUF, llama.cpp, llama-server, vLLM, OpenAI-compatible APIs, Gemini API integration patterns, local/cloud LLM deployment, model fallback, token/usage tracking, context packing, history compression

AI/ML & Data Science : PyTorch, TensorFlow, TFLite, ONNX, scikit-learn, pandas, NumPy, Darts, XGBoost, AdaBoost, ANFIS, Random Forest, Logistic Regression, SVR, MLP, LSTM, CNN-LSTM, Transformers, spatio-temporal Transformers, DLinear, YOLOv5, ResNet50, YAMNet, DeepSORT, OCR, OpenCV, feature engineering, EDA, missing-data handling, model evaluation, benchmarking, quantization, INT4/INT8

Edge AI, GPU & Accelerators : CUDA, ROCm, Vulkan, GStreamer, ROS2, embedded Linux, batch/streaming inference, NVIDIA Jetson/Orin, NVIDIA Ampere-class edge GPUs, AMD Zen x86 APUs, Radeon iGPU, Zynq UltraScale+ MPSoC/Kria, Hailo, RK3588/RKNN, Coral TPU, Raspberry Pi, Arm SBCs

Production Engineering : Python, JavaScript, TypeScript, C/C++, SQL, KQL, FastAPI, Starlette, REST APIs, SSE streaming, MQTT, Docker, Kubernetes/GKE/AKS, Cloud Run concepts, Git/GitHub Actions, GitLab, Jenkins, CI/CD, Linux, Bash/Shell, SQLite, SQLite3, Pydantic, Streamlit, monitoring, tracing, deployment debugging, release management, model packaging

Data Engineering, Cloud & Analytics : Microsoft Fabric, Azure Foundry concepts, Azure DevOps, Azure Data Factory, Dataflow Gen2, ADLS Gen2, Synapse, Power BI, Databricks, PySpark, T-SQL, BigQuery, Vertex AI concepts, AWS S3, AWS RDS, AWS EC2, AWS Lambda, AWS Trainium, SageMaker/Bedrock concepts, Boto3, GCS, Azure Blob, Cloudflare R2/D1, Grafana, Prometheus, Tableau

Customer-Facing Delivery : live demos, forward-deployed engineering, architecture walkthroughs, production-readiness reviews, benchmark reports, technical training, deployment guides, product videos, demo scripts, social-media technical demos, whitepapers, FAQs, POC scoping, MVP planning, QA notes, field feedback, sales/FAE support, client requirement translation, Jira, Confluence, Android Studio, Xcode

EDUCATION

- **M.Eng. Electrical & Computer Engineering** – *University of Victoria* 2021–2024
- **B.Eng. Computer Science** – *Amity University* 2017–2021

PUBLICATIONS

- **FUSTT[†]** – Spatio-temporal transformer for multivariate long-term forecasting of marine stressors, sponsored by NRC and published through IEEE/NRC; focused on Blue Carbon marine ecosystems, sensor forecasting, and sustainable environmental monitoring.
- **Concrete Characteristics Evaluation[†]** – Machine learning prediction of concrete physicochemical properties using classical ML and ensemble methods for sustainability-focused material analysis and carbon-emission reduction.
- **Human Activity Recognition[†]** – Evaluated human activity detection using wearable sensor data, LSTM-based sequence modeling, preprocessing pipelines, and model-performance analysis.

SELECTED AI ENGINEERING PROJECTS

- **Kaptain Core Platform[†]** – Full-stack AI-agent control and development platform with model routing, session history, governed tool execution, MCP-style tool exposure, ADK-compatible runtime paths, code graph indexing, local runtime setup, and Blackbox/God's Eye observability.

- **COMPIO[†]** – FastAPI/RAG market-intelligence and hybrid AI benchmarking platform using API/web ingestion, Yahoo Finance/news data, vector search, S3/object storage, local/cloud LLMs, and structured report generation.
- **SVSI / VaultFS Intelligence** – Secure local-first RAG interface for querying sensitive enterprise files with local LLMs, vector databases, privacy-aware retrieval, optional cloud offload, and customer-controlled data boundaries.
- **HybridTrax / Cognifix Edge AI Workflows** – Incident-response and robotics-oriented Edge AI demos combining BLE/location triggers, Python watcher services, ROS2 actions, computer vision, speech/audio recognition, TTS, REST services, and local AI agents.
- **Imagilect Computer Vision Retrieval** – Video and image intelligence features using YOLOv5 object detection, DeepSORT tracking, OCR, ResNet50 classification, OpenCV, GStreamer, and structured storage for search and scene analysis.
- **Fraud Detection ML[†]** – End-to-end classification pipeline using Logistic Regression and Random Forest with class-imbalance handling, confusion-matrix analysis, precision, recall, and F1 evaluation.
- **Stock Forecasting[†]** – Time-series forecasting workflow comparing Linear Regression, LSTM, and XGBoost models with data preparation, feature engineering, prediction visualization, and app-based outputs.